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	Section				Sheet no./rev. 1	
	Calc. by J. GILMAN	Date 14/07/2020	Chk'd by P. STEWART	Date	App'd by	Date

HSS COLUMN – CUTOUT PLATE WIDE-FLANGE BEAM MOMENT CONNECTION (AISC360-10)

In accordance with AISC360-10 and the LRFD method

Tedds calculation version 1.0.00

Connection geometry

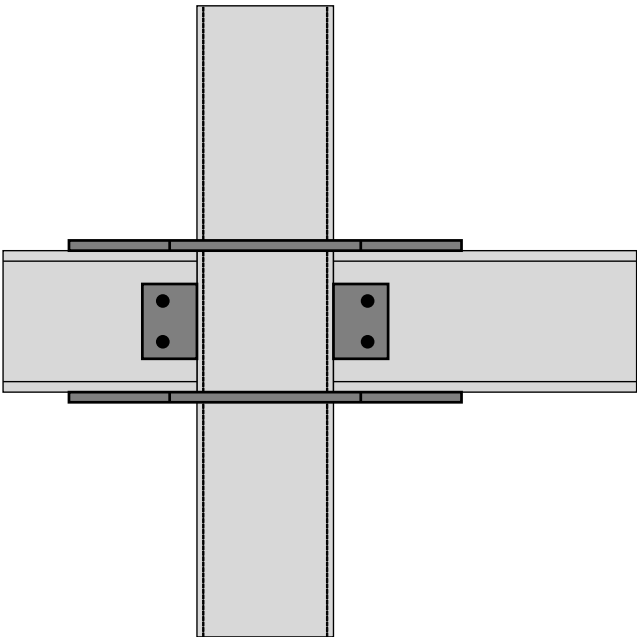
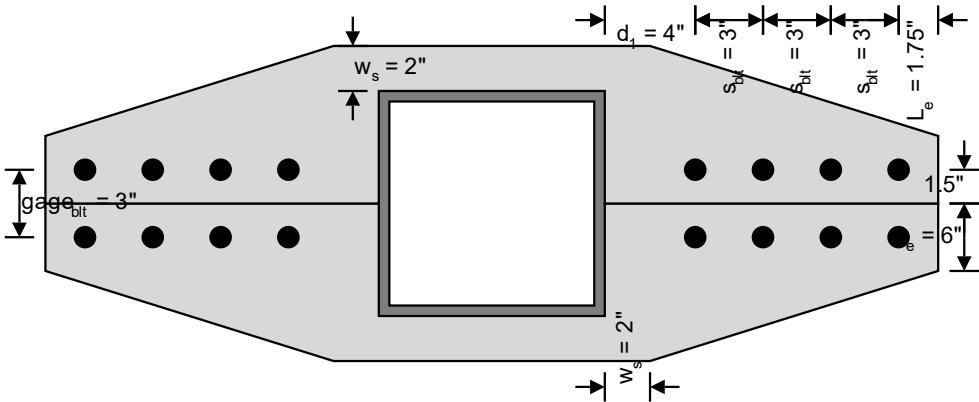


Plate geometry



Input Summary

Loading

Beam moment
 $M_r = 209 \text{ kip_ft}$

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Axial load on the cut-out plate

$$P_{r_bf} = M_r / (d_{bm} + t_{plt}) = \mathbf{224.933 \text{ kips}}$$

Cut-out plate

Thickness of the cut-out plate

$$t_{plt} = \mathbf{0.75 \text{ in}}$$

Width of the plate at the cut-out

$$w_s = \mathbf{2 \text{ in}}$$

Width of the plate at the end

$$w_e = \mathbf{6 \text{ in}}$$

Edge distance at the end of the plate

$$L_e = \mathbf{1.75 \text{ in}}$$

Plate minimum yield strength

$$F_{y_plt} = \mathbf{36 \text{ ksi}}$$

Plate tensile strength

$$F_{u_plt} = \mathbf{58 \text{ ksi}}$$

Column size

$$\mathbf{HSS 10 \times 0.500 : ASTM A500grC}$$

Column minimum yield strength

$$F_{y_col} = \mathbf{46 \text{ ksi}}$$

Column tensile strength

$$F_{u_col} = \mathbf{62 \text{ ksi}}$$

Width of the HSS column

$$B = D_{o_col} = \mathbf{10 \text{ in}}$$

Depth of the HSS column

$$H = D_{o_col} = \mathbf{10 \text{ in}}$$

Design thickness of the HSS column

$$t = t_{col} = \mathbf{0.465 \text{ in}}$$

Outside corner radius of HSS column

$$k = 1.5 \times t = \mathbf{0.698 \text{ in}}$$

Beam size

$$\mathbf{W 10 \times 68 : ASTM A992}$$

Beam minimum yield strength

$$F_{y_bm} = \mathbf{50 \text{ ksi}}$$

Beam tensile strength

$$F_{u_bm} = \mathbf{65 \text{ ksi}}$$

Beam setback (support to beam end)

$$S_{bm} = \mathbf{0 \text{ in}}$$

Bolts

$$\mathbf{A325N - STD - 1 \text{ Dia.}}$$

Number of bolts per row

$$N_{blt} = \mathbf{4}$$

Bolt gage

$$gage_{blt} = \mathbf{3 \text{ in}}$$

Bolt spacing

$$s_{blt} = \mathbf{3 \text{ in}}$$

Distance to the first row of bolts

$$d_1 = \mathbf{4 \text{ in}}$$

Spacing between bolts within a row

$$S_{blt} = \mathbf{3 \text{ in}}$$

Compression in the plate at the cut-out per AISC 360-10 Section E3

Width-to-thickness ratio of plate projection

$$ws_t = w_s / t_{plt} = \mathbf{2.67}$$

Width-to-thickness compact limit (Table B4.1a - Case 3)

$$b_t_{Lim} = 0.45 * \text{SQRT}(E_{steel} / F_{y_plt}) = \mathbf{12.77}$$

Critical buckling stress (equal to yield stress for a nonslender plate)

$$F_{cr} = F_{y_plt} = \mathbf{36.00 \text{ ksi}}$$

Resistance factor for compression

$$\phi_c = \mathbf{0.90}$$

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Design compression strength

$$P_{c_c} = P_{n_c} / \phi_c = 120.0 \text{ kips}$$

FAIL - The required compression strength exceeds the allowable compression strength.

Tensile yielding on the plate width per AISC 360-10 Section D2(a)

Gross area of the plate

$$A_g = t_{plt} * (B + 2 * w_s) = 10.50 \text{ in}^2$$

Resistance factor for tensile yielding

$$\phi_{ty} = 0.90$$

Design tensile yielding strength

$$T_{c_ty} = T_{n_ty} / \phi_{ty} = 420.0 \text{ kips}$$

PASS - The allowable tensile yielding strength exceeds the required tension strength.

Tensile rupture on the plate width per AISC 360-10 Section D2(b)

Net area of the plate

$$A_n = A_g - t_{plt} * 2 * (d_{hole_blt} + l_{damage}) = 8.81 \text{ in}^2$$

Resistance factor for tensile rupture

$$\phi_{tr} = 0.75$$

Design tensile rupture strength

$$T_{c_tr} = T_{n_tr} / \phi_{tr} = 681.4 \text{ kips}$$

PASS - The allowable tensile rupture strength exceeds the required tension strength.

Design Checks

Bolt shear per AISC 360-10 Section J3.6

Bolt designation

$$\text{Designation_blt} = \text{"A325N"}$$

Nominal shear strength (stress) of the bolt

$$F_{nv_blt} = 54 \text{ ksi}$$

Nominal shear strength of one bolt (Eq. J3-1)

$$r_{n_blt} = F_{nv_blt} * A_{nom_blt} = 42.41 \text{ kips}$$

Resistance factor

$$\phi_{bv} = 0.750$$

Design bolt shear strength

$$r_{c_blt} = r_{n_blt} * \phi_{bv} = 31.8 \text{ kips}$$

Design shear strength for bolt group

$$R_{c_blts} = r_{c_blt} * 2 * N_{blt} = 254.47 \text{ kips}$$

Required shear strength

$$P_{r_bf} = 224.9 \text{ kips}$$

PASS - The design shear strength of the bolt group exceeds the required shear strength.

Bolt bearing per AISC 360-10 Section J3.10

Nominal hole dimension

$$d_{hole_blt} = 1.063 \text{ in}$$

Nominal slot dimension

$$l_{hole_blt} = 1.063 \text{ in}$$

Clear distances to the edge of the plate

$$l_{c_edge} = L_e - l_{hole_blt} / 2 = 1.219 \text{ in}$$

Clear distance between bolt holes

$$l_{c_hole} = S_{blt} - l_{hole_blt} = 1.937 \text{ in}$$

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Bearing in the plate

Nominal bearing strength for edge bolt kips	$r_{n_ebr_plt} = \text{MIN}(1.2 * l_{c_edge} * t_{plt} * F_{u_plt}, 2.4 * dia_blt * t_{plt} * F_{u_plt}) = \mathbf{63.61}$
Nominal bearing strength for interior bolt kips	$r_{n_hbr_plt} = \text{MIN}(1.2 * l_{c_hole} * t_{plt} * F_{u_plt}, 2.4 * dia_blt * t_{plt} * F_{u_plt}) = \mathbf{101.11}$
Nominal bearing strength for bolt group	$R_{n_br_plt} = 2 * r_{n_ebr_plt} + 2 * (N_{blt} - 1) * r_{n_hbr_plt} = \mathbf{733.88}$ kips
Resistance factor for bolt bearing	$\phi_{bb} = \mathbf{2.00}$
Design bearing strength for bolt group	$R_{c_br_plt} = \phi_{bb} * R_{n_br_plt} = \mathbf{1467.8}$ kips

PASS - The design bearing strength of the plate exceeds the required shear strength.

Bearing in the beam

Nominal bearing strength for edge bolt 120.12 kips	$r_{n_ebr_bm} = \text{MIN}(1.2 * l_{ce_bm} * t_{f_bm} * F_{u_bm}, 2.4 * dia_blt * t_{f_bm} * F_{u_bm}) =$
Nominal bearing strength for interior bolt 116.34 kips	$r_{n_hbr_bm} = \text{MIN}(1.2 * l_{ch_bm} * t_{f_bm} * F_{u_bm}, 2.4 * dia_blt * t_{f_bm} * F_{u_bm}) =$
Nominal bearing strength for bolt group	$R_{n_br_bm} = 2 * r_{n_ebr_bm} + 2 * (N_{blt} - 1) * r_{n_hbr_bm} = \mathbf{938.26}$ kips
Resistance factor for bolt bearing	$\phi_{bb} = \mathbf{2.00}$
Design bearing strength for bolt group	$R_{c_br_bm} = \phi_{bb} * R_{n_br_bm} = \mathbf{1876.5}$ kips

PASS - The design bearing strength of the beam web exceeds the required shear strength.

Block shear in the plate per AISC 360-10 Section J4.3

Gross length subject to shear	$L_{gv_BS} = L_e + (N_{blt} - 1) * S_{blt} = \mathbf{10.750}$ in
Gross area subject to shear	$A_{gv_BS} = L_{gv_BS} * t_{plt} = \mathbf{8.06}$ in ²
Net area subject to shear	$A_{nv_BS} = A_{gv_BS} - ((N_{blt} - 0.5) * (l_{hole_blt} + l_{damage})) * t_{plt}$ $A_{nv_BS} = \mathbf{5.11}$ in ²
Net area subject to tension	$A_{nt_BS} = (g_{age_blt} - (d_{hole_blt} + l_{damage})) * t_{plt}$ $A_{nt_BS} = \mathbf{1.41}$ in ²
Reduction factor for nonuniform stress	$U_{bs} = \mathbf{1.000}$ for one row of bolts (See Commentary on Section J4.3)
Nominal block shear strength: Shear rupture on the net section	$R_{n_bs_sr} = 0.6 * F_{u_plt} * A_{nv_BS} + U_{bs} * F_{u_plt} * A_{nt_BS} = \mathbf{259.3}$ kips
Nominal block shear strength: Shear yield on the gross section	$R_{n_bs_sy} = 0.6 * F_{y_plt} * A_{gv_BS} + U_{bs} * F_{u_plt} * A_{nt_BS} = \mathbf{255.7}$ kips
Nominal block shear strength	$R_{n_bs} = \text{MIN}(R_{n_bs_sr}, R_{n_bs_sy}) = \mathbf{255.7}$ kips

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Resistance factor	$\phi_{bs} = 2.00$
Design block shear strength	$R_{c_bs} = \phi_{bs} * R_{n_bs} = 511.4$ kips
Block shear in the beam flange per AISC 360-10 Section J4.3	
Gross length subject to shear	$L_{gv_BS_BM} = L_{ep} + (N_{blt}-1)*s_{blt} = 13.000$ in
Gross area subject to shear	$A_{gv_BS_BM} = L_{gv_BS_BM} * t_{f_bm} = 10.01$ in ²
Net area subject to shear	$A_{nv_BS_BM} = A_{gv_BS_BM} - ((N_{blt}-0.5)*(l_{hole_blt} + l_{damage}))*t_{f_bm}$ $A_{nv_BS_BM} = 6.98$ in ²
Net area subject to tension	$A_{nt_BS_BM} = (2*L_{et} - (d_{hole_blt} + l_{damage}))*t_{f_bm}$ $A_{nt_BS_BM} = 4.60$ in ²
Reduction factor for nonuniform stress	$U_{bs} = 1.000$ for one row of bolts (See Commentary on Section J4.3)
Nominal block shear strength: Shear rupture on the net section	
	$R_{n_bs_sr_bm} = 0.6*F_{u_plt}*A_{nv_BS_BM} + U_{bs}*F_{u_plt}*A_{nt_BS_BM} = 509.6$ kips
Nominal block shear strength: Shear yield on the gross section	
	$R_{n_bs_sy_bm} = 0.6*F_{y_plt}*A_{gv_BS_BM} + U_{bs}*F_{u_plt}*A_{nt_BS_BM} = 483.0$ kips
Nominal block shear strength	$R_{n_bs_bm} = \text{MIN}(R_{n_bs_sr_bm}, R_{n_bs_sy_bm}) = 483.0$ kips
Resistance factor	$\phi_{bs} = 2.00$
Design block shear strength	$R_{c_bs_bm} = \phi_{bs} * R_{n_bs_bm} = 966.1$ kips

PASS - The design block shear strength exceeds the required shear strength.